Figure 1.1. Integrated Concept Map: Foundations and Historical Development of Trauma Neurobiology

Chapter topics and a reading framework

- 1.1 Scope of neuroscience
- 1.2 Meaning of trauma in biological research
- 1.3 History of stress science
- 1.4 Early lesion and behavioural studies
- 1.5 Rise of psychophysiology
- 1.6 Development of neuroendocrinology
- 1.7 Growth of brain imaging
- 1.8 Translation from laboratory to clinic
- 1.9 Limits of biological reductionism
- 1.10 Integrative contemporary models

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 1, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 1.2. Evidence-to-Practice Pathway: Foundations and Historical Development of Trauma Neurobiology



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 1, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 2.1. Integrated Concept Map: Functional Neuroanatomy of Threat, Memory, and Regulation

Chapter topics and a reading framework

- 2.1 Orientation to the nervous system
- 2.2 Cerebral cortex and subcortex
- 2.3 Amygdala organisation
- 2.4 Hippocampal formation
- 2.5 Prefrontal cortical systems
- 2.6 Insula and interoception
- 2.7 Anterior cingulate cortex
- 2.8 Brainstem and periaqueductal grey
- 2.9 Cerebellar contributions
- 2.10 Large-scale brain networks

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 2, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 2.2. Evidence-to-Practice Pathway: Functional Neuroanatomy of Threat, Memory, and Regulation



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 2, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 3.1. Integrated Concept Map: Brain Development Across the Lifespan and Developmental Timing of Trauma

Chapter topics and a reading framework

- 3.1 Prenatal neural development
- 3.2 Infancy and early caregiving
- 3.3 Childhood synaptic refinement
- 3.4 Adolescent maturation
- 3.5 Adult plasticity
- 3.6 Ageing and cumulative stress
- 3.7 Sensitive periods
- 3.8 Attachment-related regulation
- 3.9 Developmental adversity
- 3.10 Lifespan recovery

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 3, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 3.2. Evidence-to-Practice Pathway: Brain Development Across the Lifespan and Developmental Timing of Trauma



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 3, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 4.1. Integrated Concept Map: Cellular Neuroscience: Neurons, Glia, Myelin, and Neural Tissue Responses

Chapter topics and a reading framework

- 4.1 Cell theory and nervous tissue
- 4.2 Neuronal structure
- 4.3 Membrane potentials
- 4.4 Action potentials
- 4.5 Astrocyte functions
- 4.6 Microglial surveillance
- 4.7 Oligodendrocytes and myelin
- 4.8 Blood-brain barrier
- 4.9 Cellular stress responses
- 4.10 Cellular repair and adaptation

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 4, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 4.2. Evidence-to-Practice Pathway: Cellular Neuroscience: Neurons, Glia, Myelin, and Neural Tissue Responses

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 4, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 5.1. Integrated Concept Map: Molecular Neuroscience and Synaptic Transmission in Traumatic Stress

Chapter topics and a reading framework

- 5.1 Molecular organisation of synapses
- 5.2 Electrical and chemical signalling
- 5.3 Vesicle release
- 5.4 Receptor families
- 5.5 Second-messenger systems
- 5.6 Gene transcription
- 5.7 Protein synthesis
- 5.8 Oxidative stress
- 5.9 Mitochondrial regulation
- 5.10 Molecular adaptation after stress

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 5, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 5.2. Evidence-to-Practice Pathway: Molecular Neuroscience and Synaptic Transmission in Traumatic Stress

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 5, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 6.1. Integrated Concept Map: Neurotransmitters, Neuromodulators, and Neuroendocrine Signalling

Chapter topics and a reading framework

- 6.1 Glutamate signalling
- 6.2 GABA inhibition
- 6.3 Noradrenergic arousal
- 6.4 Serotonergic modulation
- 6.5 Dopamine and salience
- 6.6 Acetylcholine and attention
- 6.7 Endogenous opioids
- 6.8 Endocannabinoid regulation
- 6.9 Neuropeptides
- 6.10 Hormone-brain communication

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 6, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 6.2. Evidence-to-Practice Pathway: Neurotransmitters, Neuromodulators, and Neuroendocrine Signalling

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 6, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 7.1. Integrated Concept Map: Neuroplasticity, Learning, Memory, and Reconsolidation

Chapter topics and a reading framework

- 7.1 Forms of neuroplasticity
- 7.2 Long-term potentiation
- 7.3 Long-term depression
- 7.4 Memory systems
- 7.5 Encoding and consolidation
- 7.6 Retrieval and updating
- 7.7 Reconsolidation
- 7.8 Extinction learning
- 7.9 Habit and procedural learning
- 7.10 Experience-dependent recovery

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 7, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 7.2. Evidence-to-Practice Pathway: Neuroplasticity, Learning, Memory, and Reconsolidation

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 7, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 8.1. Integrated Concept Map: Stress Physiology: HPA Axis, Autonomic Regulation, and Allostasis

Chapter topics and a reading framework

- 8.1 Definition of stress
- 8.2 Acute stress response
- 8.3 HPA axis sequence
- 8.4 Glucocorticoid feedback
- 8.5 Sympathetic activation
- 8.6 Parasympathetic regulation
- 8.7 Cardiorespiratory responses
- 8.8 Allostatic load
- 8.9 Sleep and circadian regulation
- 8.10 Recovery and physiological flexibility

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 8, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 8.2. Evidence-to-Practice Pathway: Stress Physiology: HPA Axis, Autonomic Regulation, and Allostasis

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 8, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 9.1. Integrated Concept Map: Genetics, Epigenetics, and Gene-Environment Interplay

Chapter topics and a reading framework

- 9.1 Genetic variation
- 9.2 Heritability and its limits
- 9.3 Candidate gene history
- 9.4 Genome-wide association
- 9.5 Gene-environment interplay
- 9.6 DNA methylation
- 9.7 Histone modification
- 9.8 Non-coding rna
- 9.9 Intergenerational questions
- 9.10 Ethical genomics and precision research

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 9, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 9.2. Evidence-to-Practice Pathway: Genetics, Epigenetics, and Gene-Environment Interplay



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 9, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 10.1. Integrated Concept Map: Threat Learning, Fear Conditioning, and Traumatic Memory

Chapter topics and a reading framework

10.1 Pavlovian conditioning

10.2 Unconditioned and conditioned responses

10.3 Acquisition of threat learning

10.4 Context conditioning

10.5 Fear generalisation

10.6 Avoidance learning

10.7 Extinction and renewal

10.8 Prediction error

10.9 Intrusive memory

10.10 Clinical translation of learning science

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 10, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 10.2. Evidence-to-Practice Pathway: Threat Learning, Fear Conditioning, and Traumatic Memory



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 10, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 11.1. Integrated Concept Map: Emotion Regulation, Interoception, and Dissociation

Chapter topics and a reading framework

- 11.1 Emotion generation
- 11.2 Regulatory strategies
- 11.3 Cognitive reappraisal
- 11.4 Attention and inhibition
- 11.5 Interoceptive processing
- 11.6 Body awareness
- 11.7 Depersonalisation
- 11.8 Derealisation
- 11.9 State-dependent responding
- 11.10 Assessment of dissociative phenomena

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 11, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 11.2. Evidence-to-Practice Pathway: Emotion Regulation, Interoception, and Dissociation



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 11, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 12.1. Integrated Concept Map: Neurobiology of Post-Traumatic Stress Disorder

Chapter topics and a reading framework

- 12.1 Diagnostic concepts and symptom clusters
- 12.2 Threat circuitry findings
- 12.3 Memory and contextual processing
- 12.4 Prefrontal regulation
- 12.5 Arousal and sleep
- 12.6 Endocrine findings
- 12.7 Immune findings
- 12.8 Structural imaging findings
- 12.9 Functional connectivity
- 12.10 Heterogeneity and comorbidity

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 12, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 12.2. Evidence-to-Practice Pathway: Neurobiology of Post-Traumatic Stress Disorder



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 12, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 13.1. Integrated Concept Map: Complex Trauma, Developmental Trauma, and Childhood Adversity

Chapter topics and a reading framework

13.1 Definition and boundaries of complex trauma

13.2 Chronic interpersonal threat

13.3 Developmental timing

13.4 Attachment and co-regulation

13.5 Self-organisation difficulties

13.6 Cognition and executive function

13.7 Somatic and pain pathways

13.8 Social neurobiology

13.9 Institutional and structural adversity

13.10 Developmentally responsive care

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 13, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 13.2. Evidence-to-Practice Pathway: Complex Trauma, Developmental Trauma, and Childhood Adversity



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 13, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 14.1. Integrated Concept Map: Neurobiology of Resilience, Recovery, and Adaptive Regulation

Chapter topics and a reading framework

- 14.1 Definitions of resilience
- 14.2 Dynamic adaptation
- 14.3 Protective relationships
- 14.4 Cognitive flexibility
- 14.5 Reward and motivation
- 14.6 Stress inoculation debates
- 14.7 Sleep and physical activity
- 14.8 Social connection
- 14.9 Recovery trajectories
- 14.10 Measurement and intervention research

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 14, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 14.2. Evidence-to-Practice Pathway: Neurobiology of Resilience, Recovery, and Adaptive Regulation

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 14, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 15.1. Integrated Concept Map: Neuroinflammation and Psychoneuroimmunology of Trauma

Chapter topics and a reading framework

- 15.1 Immune system foundations
- 15.2 Innate and adaptive immunity
- 15.3 Cytokine signalling
- 15.4 Microglial activation
- 15.5 Brain-immune communication
- 15.6 Inflammation and mood
- 15.7 Stress-related immune change
- 15.8 Metabolic and cardiovascular links
- 15.9 Measurement of inflammatory markers
- 15.10 Anti-inflammatory intervention questions

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 15, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 15.2. Evidence-to-Practice Pathway: Neuroinflammation and Psychoneuroimmunology of Trauma

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 15, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 16.1. Integrated Concept Map: Biomarkers, Precision Neuroscience, and Individualised Prediction

Chapter topics and a reading framework

16.1 Meaning of a biomarker

16.2 Diagnostic versus prognostic markers

16.3 Measurement reliability

16.4 Multimodal biomarkers

16.5 Risk prediction

16.6 Treatment selection

16.7 Replication and validation

16.8 Fairness across populations

16.9 Clinical utility

16.10 Precision neuroscience governance

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 16, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 16.2. Evidence-to-Practice Pathway: Biomarkers, Precision Neuroscience, and Individualised Prediction



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 16, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 17.1. Integrated Concept Map: Neuroimaging, Electrophysiology, and Advanced Brain Mapping

Chapter topics and a reading framework

- 17.1 Structural MRI
- 17.2 Diffusion MRI
- 17.3 Functional MRI
- 17.4 PET imaging
- 17.5 EEG
- 17.6 MEG
- 17.7 Task design
- 17.8 Resting-state connectivity
- 17.9 Multimodal integration
- 17.10 Reproducibility and reporting

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 17, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 17.2. Evidence-to-Practice Pathway: Neuroimaging, Electrophysiology, and Advanced Brain Mapping



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 17, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 18.1. Integrated Concept Map: Neurobiological Mechanisms of Treatment and Recovery Interventions

Chapter topics and a reading framework

- 18.1 Pharmacological treatment principles
- 18.2 Fear extinction therapies
- 18.3 Cognitive therapies
- 18.4 Emdr mechanism research
- 18.5 Sleep-focused interventions
- 18.6 Exercise and behavioural medicine
- 18.7 Neuromodulation
- 18.8 Memory reconsolidation interventions
- 18.9 Combined and staged care
- 18.10 Outcome monitoring and adverse effects

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 18, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 18.2. Evidence-to-Practice Pathway: Neurobiological Mechanisms of Treatment and Recovery Interventions

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 18, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 19.1. Integrated Concept Map: Artificial Intelligence, Machine Learning, and Emerging Neurotechnologies

Chapter topics and a reading framework

- 19.1 AI foundations
- 19.2 Supervised learning
- 19.3 Unsupervised learning
- 19.4 Deep learning
- 19.5 Neuroimaging classification
- 19.6 Digital phenotyping
- 19.7 Wearable sensing
- 19.8 Brain-computer interfaces
- 19.9 Explainability and bias
- 19.10 Responsible innovation

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 19, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 19.2. Evidence-to-Practice Pathway: Artificial Intelligence, Machine Learning, and Emerging Neurotechnologies



Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 19, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 20.1. Integrated Concept Map: Advanced Research Methods, Translation, Ethics, Policy, and Future Directions

Chapter topics and a reading framework

- 20.1 Research questions and theory
- 20.2 Experimental design
- 20.3 Observational design
- 20.4 Longitudinal research
- 20.5 Qualitative inquiry
- 20.6 Mixed-methods research
- 20.7 Systematic review and meta-analysis
- 20.8 Open science and reproducibility
- 20.9 Implementation and policy research
- 20.10 Doctoral research agenda

Read across levels

Neural • endocrine • immune • cognitive • developmental • social

For each topic: define the construct, examine the method, interpret uncertainty, and consider the decision or application.

Boxes index the chapter's existing section headings. Position is for navigation, not anatomical location or causal order. No connecting arrows are used because the outline alone cannot establish biological pathways.

Source: Chapter 20, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.

Figure 20.2. Evidence-to-Practice Pathway: Advanced Research Methods, Translation, Ethics, Policy, and Future Directions

1 Define the question and context

Specify the exposure, population, timing, and decision. Distinguish the question from the explanation.

↓

2 Specify the process and evidence

Identify the relevant biological level and the method used to study it. Separate observation from inference.

↓

3 Assess outcomes and alternatives

Consider symptoms, function, safety, context, and competing explanations. Group findings do not determine an individual diagnosis.

↓

4 Select a proportionate action

Relate evidence to assessment, education, supervised practice, or research. Record limits and responsibilities.

↓

5 Monitor and review

Review outcomes, uncertainty, adverse effects, and new information. Revise the interpretation when warranted.

Arrows show an educational decision sequence, not biological causation or a validated clinical algorithm. Apply this framework to the chapter topics; it does not prescribe treatment or establish diagnostic accuracy.

Source: Chapter 20, section outline and existing discussion of evidence, interpretation, application, and review. Original editorial schematic; no empirical data are plotted. Subject-author review remains required.